Arithmetic and complexity

Minimal asymptotic rank of the small Coppersmith–Winograd tensor

Difficulty: extreme **Importance:** broadly interesting

Topic: asymptotic tensor algorithms

Status: open; admitted after a bounded literature search found no resolution.

Context and notation

Over $\mathbb{C}$, the tensor rank $R(T)$ is the minimum number of pure tensors in an exact sum for T.

Problem statement

Let e_0, e_1, e_2 be the standard basis of $\mathbb{C}^3$ and set

$$T = \sum_{i=1}^{2} (e_0 \otimes e_i \otimes e_i + e_i \otimes e_0 \otimes e_i + e_i \otimes e_i \otimes e_0).$$

For a tensor S, define $\widetilde{R}(S) = \lim_{k\to\infty} R(S^{\otimes k})^{1/k}$, grouping corresponding factors when taking powers. Is $\widetilde{R}(T) = 3$?

Why it matters

This explicitly studied tensor offers a route to exponent two for matrix multiplication. It is not counted separately for other values of its size parameter.

References and status

Bläser, 2013, Problem 9.8. Alman–Li, 2026, Theorems 1.2–1.3, relates the target to $\omega = 2$ and establishes the partial upper bound $\widetilde{R}(T) < 3.931$. Searches for “small Coppersmith Winograd asymptotic rank 2026 3” located that improvement, not equality. Border-rank results for a fixed tensor power do not alone determine this limit. **Admitted: no resolution located.**